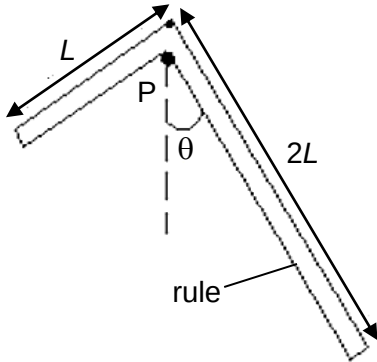


8	A right-angle rule hangs at rest from a peg P as shown below. The rule is uniform in density and cross-sectional area. One arm is of length L while the other arm is of length $2L$.  What is the angle θ at which the rule will hang in equilibrium?								
	<table> <tr> <td data-bbox="197 591 261 624">A</td><td data-bbox="261 591 517 624">8°</td><td data-bbox="517 591 595 624">B</td><td data-bbox="595 591 823 624">14°</td><td data-bbox="823 591 914 624">C</td><td data-bbox="914 591 1158 624">42°</td><td data-bbox="1158 591 1233 624">D</td><td data-bbox="1233 591 1444 624">76°</td></tr> </table>	A	8°	B	14°	C	42°	D	76°
A	8°	B	14°	C	42°	D	76°		
L3	Answer: B Let m be the mass of the arm of length L. Since rule is of uniform density and uniform cross-sectional area, the other arm of length $2L$ will have mass $2m$. Take P as the pivot, Total anticlockwise moments = Total clockwise moments $(mg) \left[\frac{L}{2} (\cos \theta) \right] = (2mg) \left[\frac{2L}{2} (\sin \theta) \right]$ $[(\cos \theta)] = (2)[2(\sin \theta)]$ $\tan \theta = \frac{1}{4}$ $\theta = 14^\circ$								

9	A car of mass 1200 kg travels along a horizontal road at a speed of 10 m s^{-1}. At the time it just
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